

TOURS VAL DE LOIRE AP, France

WMO: 072400

Lat: 47.4444N

Lon: 0.7275E

Elev: 109

StdP: 100.02

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-4.3	-2.7	-7.9	1.9	-2.9	-5.8	2.3	-1.4	12.1	11.0	10.7	10.1	3.3	50	0.524

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.0	31.7	20.3	29.5	19.5	27.6	18.8	21.5	29.1	20.5	27.4	19.6	25.8	3.4	100

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
19.1	14.0	24.2	18.1	13.2	22.9	17.2	12.4	22.0	62.9	29.2	59.2	27.4	56.1	25.9	

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.4	8.3	7.3	-6.6	35.1	2.7	2.0	-8.5	36.5	-10.1	37.7	-11.6	38.8	-13.5	40.2		
			-7.0	23.3	2.7	1.2	-9.0	24.1	-10.6	24.8	-12.1	25.5	-14.1	26.4		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	12.1	4.9	5.6	8.4	10.8	14.3	17.9	20.1	19.9	16.5	13.0	8.2	5.4	(6)
		DBStd	6.41	3.94	3.74	3.33	3.27	3.17	3.42	3.05	3.13	2.95	3.42	3.51	3.96	
		HDD10.0	623	159	126	72	28	3	0	0	0	12	74	148		
		HDD18.3	2463	415	356	309	227	130	49	16	17	71	167	305	401	
		CDD10.0	1395	3	3	21	52	137	237	312	306	195	106	19	5	
		CDD18.3	193	0	0	0	1	5	35	70	65	16	2	0	0	
		CDH23.3	1873	0	0	0	12	64	358	666	613	148	12	0	0	
		CDH26.7	616	0	0	0	1	8	116	228	229	33	1	0	0	
(14)	Wind	WSAvg	3.7	4.1	4.0	4.0	3.8	3.6	3.4	3.4	3.2	3.3	3.4	3.7	4.0	(14)
(15)	Precipitation	PrecAvg	674	64	54	52	52	61	52	49	45	54	63	70	63	(15)
(16)		PrecMax	930	137	145	122	145	138	135	111	170	174	152	168	126	(16)
(17)		PrecMin	519	7	5	3	7	11	0	5	1	10	3	12	12	(17)
(18)		PrecStd	107	32	32	29	30	32	30	27	34	35	39	38	31	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	13.6	16.2	20.8	25.0	27.7	33.7	34.3	35.4	30.1	24.7	17.8	14.2	(19)
(20)			MCWB	11.8	10.8	13.0	16.1	18.9	21.8	20.5	21.8	18.6	17.6	14.5	12.4	(20)
(21)		2%	DB	12.5	13.5	18.0	21.9	25.1	29.7	31.4	31.9	27.1	22.0	15.8	13.1	(21)
(22)			MCWB	11.1	10.0	12.0	14.7	17.4	19.8	20.2	20.2	18.1	16.3	13.6	11.5	(22)
(23)		5%	DB	11.4	11.9	15.6	19.3	23.0	27.3	29.2	29.1	24.8	19.7	14.4	12.1	(23)
(24)			MCWB	10.0	9.7	11.1	13.2	16.6	19.1	19.2	19.5	17.1	15.7	12.6	10.8	(24)
(25)		10%	DB	10.2	10.8	13.9	16.9	20.9	24.9	27.2	26.7	22.5	18.1	13.1	11.0	(25)
(26)			MCWB	9.0	9.1	10.4	11.9	15.4	18.1	18.6	18.5	16.1	15.0	11.5	9.8	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	12.4	12.0	14.0	17.1	20.1	22.7	22.3	22.9	20.4	18.4	15.5	13.1	(27)
(28)			MCDB	13.1	14.5	19.0	23.4	26.3	30.1	30.6	31.4	26.4	22.8	16.9	13.8	(28)
(29)		2%	WB	11.3	10.9	12.7	15.2	18.4	21.1	21.1	21.4	19.1	17.2	14.0	11.9	(29)
(30)			MCDB	12.3	12.4	16.3	20.8	23.5	27.9	29.0	29.1	24.7	20.5	15.6	12.9	(30)
(31)		5%	WB	10.2	10.1	11.8	13.7	17.0	19.8	20.1	20.3	18.0	16.3	12.7	11.0	(31)
(32)			MCDB	11.2	11.5	14.6	18.3	22.0	25.7	27.3	27.2	23.4	19.1	14.1	12.0	(32)
(33)		10%	WB	9.1	9.2	10.9	12.4	15.8	18.6	19.2	19.3	17.0	15.4	11.7	10.0	(33)
(34)			MCDB	10.1	10.7	13.4	16.2	20.0	24.1	25.6	25.1	21.2	17.6	12.9	10.9	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	5.0	6.5	8.4	9.5	9.7	10.6	11.0	11.1	10.4	8.1	6.1	5.2	(35)
(36)			MCDBR	5.5	8.1	11.5	13.3	13.3	14.3	14.5	15.0	13.7	10.3	6.9	5.7	(36)
(37)		5% WB	MCWBR	4.6	5.7	6.8	7.2	7.0	6.4	5.4	5.9	6.0	5.7	5.0	4.7	(37)
(38)			MCDWR	5.2	7.0	9.4	12.0	12.1	12.9	12.6	12.6	11.5	9.0	6.5	5.3	(38)
(39)	Clear-Sky Solar Irradiance	taub		0.322	0.342	0.370	0.396	0.398	0.411	0.403	0.392	0.365	0.361	0.334	0.318	(40)
(41)			taud	2.445	2.386	2.309	2.238	2.277	2.278	2.295	2.344	2.400	2.421	2.443	2.425	(41)
(42)		Ebn at Noon		758	817	845	855	866	854	855	850	842	783	733	712	(42)
(43)			Edh at Noon	67	87	109	129	129	129	125	115	99	83	67	61	(43)
(44)	All-Sky Solar Radiation	RadAvg		1.04	1.87	3.08	4.45	5.24	5.88	5.70	5.02	3.88	2.32	1.25	0.89	(44)
(45)		RadStd		0.10	0.26	0.38	0.54	0.47	0.57	0.36	0.47	0.27	0.20	0.13	0.13	(45)

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(47)	Regional (29 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page